

Vihaan Gupta

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EDUCATION

Harvard University

May 2027

M.S. in Computer Science; GPA: 4.0/4.0

- Relevant Coursework: Distributed Systems, Human-Computer Interaction, Differential Privacy.

Harvard University

May 2027

B.A. in Computer Science and Mathematics with a secondary in Philosophy; GPA: 3.9/4.0.

- Relevant Coursework: Systems Programming, Data Structures & Algorithms, Software Engineering.
- Activities and Leadership: TA for Data Structures and Algorithms, President Tech for Social Good.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, C#, Linux Script, Dart, R, SQL.

Frameworks & Tools: Spring Boot, Kubernetes, Docker Apache Kafka, Jenkins, React, Flutter, Git.

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Amazon Web Services (AWS)

New York City, NY

- Developed a retrieval-augmented coding agent using LLMs, semantic search, and structured repository context to support automated code generation for AWS payment integration team.
- Engineered an AI-powered Slack agent that analyzes AWS partner integration dashboard alerts and generates root-cause analyses, severity assessments, and mitigation recommendations.

Founding Software Engineer

Nov 2025 – Jan 2026

Stealth Startup

Cambridge, MA

- Designed and shipped an end-to-end Voice AI agent (ElevenLabs + LLM workflows) to handle inbound calls, lead qualification, and appointment scheduling, for home service businesses.
- Architected a Node/TypeScript services and Postgres schema; implemented scheduling logic, Stripe payments, and CRM/dispatch integrations to enable real-time booking and reliable data sync.

Software Engineering Intern

June – Aug 2025

MIAX Exchange

Princeton, NJ

- Deployed a Kafka-Spring Boot microservice for real-time trade event streaming, achieving sub-millisecond message latency and ensuring end-to-end delivery guarantees under high throughput.
- Integrated kubeLint into Jenkins CI/CD to automatically scan Kubernetes config on PRs and developed linting rules tailored to compliance policies, detecting misconfigurations in 70% of PRs.

Founding Software Engineer

Mar – May 2025

As1 Social

Cambridge, MA

- Engineered 10+ features for the MVP of social media app in React Native and PostgreSQL.
- Boosted user retention by 30% by devising caption parsing and hashtag ranking for video recommendations, enabling personalised feeds based on preferences, keyword relevance and recency.

PROJECTS

Lumi | *React, RAG, REST APIs, Supabase*

Nov 2025 – Present

- Built a context-aware AI reading and writing assistant using folder-scoped retrieval-augmented generation (RAG), enabling citation-grounded drafting and editing across PDF file formats.
- Developed a scalable document ingestion and retrieval pipeline (OCR, chunking, embeddings, PostgreSQL + pgvector) capable of serving low-latency (<2s) retrieval-augmented queries.

Replicated Editor | *C++, Multi-Paxos, WebSockets, Networking*

Jan – May 2026

- Implemented a fault-tolerant collaborative text editor via a Multi-Paxos replicated log, guaranteeing consistent document state across concurrent clients while tolerating replica failures.
- Architected an end-to-end distributed architecture spanning custom RFC 6455 WebSocket networking, leader-based consensus replication, and late-join state synchronization.